VELAMMAL
INSTITUTE OF TECHNOLOGY
Approved by AICTE - New Delhi, Affiliated to Anna University - Chennai
Accredited by NATIONAL ASSESSMENT AND ACCREDITATION COUNCIL
"Velammal Knowledge Park", Chennai - Kolkatta Highway, Ponneri - 601204

AUTOMATED DRAIN CLOG REMOVAL BOT

Project Done By

- Jagadeesh M
- Bhuvaneshwaran M B
- Shane Dani Ebinesan S
- Naresh M
- Mohanapriya R S
- Dhanusha K

Mentor

Dr.B.Madhu

TABLE OF CONTENTS

1 About Manual Scavenging

2 Why Manual Scavenging?

3 Gas Analysis

4 Drain Clog Removal Bot

5 Importance

6 Government Action

7 Conclusion

Manual Scavenging : The Harsh Reality

- "Manual Scavenging" is a term used mainly in India for "manually cleaning, carrying, disposing or handling human waste in an open drain, insanitary latrine or in a septic tank".
- Hand tools such as buckets, brooms and shovels are used by Manual scavengers.
- Manual scavengers rarely have any personal protective equipment which leads to serious health issues or even death.
- The daily wage of a manual scavenger varies from 800-1500 rupees based on their work.

About Sanitation workers in India

- A total of about 5 million sanitation workers from various urban and rural areas across India involves in faecal sludge handling, latrines cleaning, drainage scavenging, etc.
- Shockingly, over 1035 people have died while undertaking hazardous cleaning of sewers and septic tanks across India since 1993.
- Even though, manual scavenging is banned in India, the practice is still commonplace with people forced into it because of rigid caste rules and lack of other livelihood options.

Why Manual Scavenging Is Still Being Practised?

- In India, most municipalities do not have the latest sewage cleaning machines and thus Manual Scavengers are forced to enter the underground sewage lines through manholes.
- Manual Scavengers are mostly uneducated and are much cheaper to hire illegally at a daily wage.
- Lack of economic alternative forces them to continue engaging in scavenging for survival
- Manual scavengers usually remove the blockage in drain which contains toxic materials including human excreta, medicinal wastes and other domestic wastes.

The Fatality Rate In India's sewer system

2018
- 76 loss of life occurred while cleaning sewer system

2019
- 133 loss of life

2020
- 35 loss of life

2021
- 66 loss of life

2022
- 84 loss of life

2023
- 49 loss of life

Challenges Faced by Manual Scavengers in India

Health hazards such as cholera, typhoid, hepatitis and various respiratory infections.

Frequent manual scavenging leads to skin cancer, asthma and even leads to premature deaths.

This occupation perpetuates social stigma, discrimination and reinforce caste based oppression.

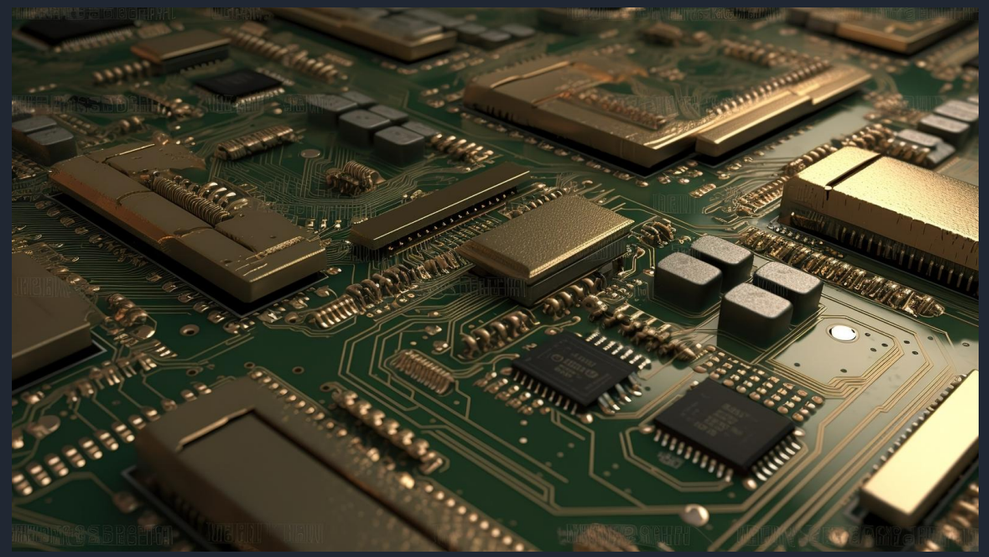
Constant exposure to filth take a toll on the mental well being causing depression, feelings of shame, low self esteem and a long term psychological trauma.

Gas Analysis In Sewer System

- The composition of gases in sewer systems can vary depending on factors such as location, time of day and waste characteristics.
- Typical percentages of gases found in sewer systems may include:
  - Methane (Ch4): 50-75%
  - Carbon dioxide(CO2): 25-5%
  - Hydrogen Sulfide(H2S): 0.1-5%
  - Nitrogen (N2): 0-5%
  - Oxygen(02): 0-1%
- Hydrogen sulfide possess the greater risk for humans.

Introduction to Automated Drain Clog Removal Bot

- An automated drain clog removal bot is a revolutionary machine designed to unclog drains efficiently and autonomously.
- To locate and clear blockages, ensuring smooth drainage systems.
- This innovative technology aims to minimize manual intervention and labor in drain cleaning processes.



Components Required for Operation

- Chasis – Framework of the Bot
- Battery or power supply unit
- Motors, wheels or tracks for locomotion and movement within drainage
- Proximity sensors, ultrasonic sensors and cameras to detect obstacles, measure distance and navigate the drainage environment
- Tools such as grippers, brushes
- Control System – Microcontroller to process sensor data
- Air Quality Sensor – Gas analysis

Importance of Automated Drain Clog Removal Bot in Improving Safety

- Automated drain clog removal bots play a crucial role in enhancing safety by reducing the need for manual intervention in hazardous drainage cleaning processes.

- With their advanced technology and efficiency, these bots minimize the risk of exposure to toxic substances and prevent workers from entering confined and dangerous spaces.

Impact of automated drain clog removal bot on the drainage industry

Increased Efficiency

The automated drain clog removal bot significantly reduces the time required to clear clogs, improving overall efficiency in the drainage industry.

Enhanced Safety

With the bot handling the riskier aspects of drain cleaning, workers are exposed to fewer hazardous conditions, leading to improved safety standards.

Boosted Productivity

Workers can focus on other important tasks while the bot effectively deals with clogs, resulting in enhanced productivity in the drainage industry.

Reliability

The consistent performance of the automated drain clog removal bot ensures that drainage systems are maintained efficiently and reliably.

Involvement of Government in Manual Scavenging

National Movement

- "Safai Karmachari Andolan (SKA), 1995" national movement was to eradicate manual scavengers for dignified occupation.

Act

- "The Prohibition of Employment as Manual Scavengers and their Rehabilitation Bill, 2012" act was to ensure prohibition of Manual scavengers.

Competition

- "Safaimitra Suraksha Challenge, 2020" competition was to ensure zero manual entry into sewers and septic tanks.

Involvement of Government in Manual Scavenging

Scheme

- "National Action for Mechanised Sanitation Ecosystem (NAMASTE), 2022" scheme was to provide Ayushman Health Cards.

Law

- The punishment for employing a manual scavenger is jail time upto 1 year as well as fine amount upto 50,000 rupees.

Compensation

- Even though around 1633 people died due to Manual Scavenging, only 948 families were given compensations by the Lok Sabha.

Investments and Success Rate

- In 2014, Supreme court's order made the government identify all those who died in sewage work since 1993 and provide 10 lakh rupees each as compensation to their families.
- Approximately, 58,000 manual scavenger have been identified and given 40,000 rupees each and engaged in skill training programs.
- For NAMASTE scheme, around 100 crores were allocated for 100% mechanisation of sewer work.
- Even though, several lives of manual scavengers were changed, still many suffer this injustice.

Inventions made for sewage cleaning in India

01

- Truck mounted Sewer suction Machine which can suck around 5000 litres of sewage water costing around 4,95,000 rupees created drastic change in the lives of Manual Scavengers.

02

- Jetting pump and blow back vacuum pump which weighs around 5400 kg can suck upto 12000 litres of sewage water costing around 8,00,000 was also efficient.

Conclusion and Key Takeaways

After exploring the impact of automated drain clog removal technology and the challenges faced by drainage workers in India, it's evident that the implementation of this technology is essential for improving safety and efficiency in the drainage industry. The key takeaways include the significance of improved working conditions, the potential reduction in occupational hazards, and the overall positive influence on the drainage system.

THANK YOU